

Harshavardhan Raja

Senior Data Engineer | Cloud & Data Platforms | 6+ Years Experience

Professional Summary

Senior Data Engineer with 6+ years of experience building and operating large-scale, production data platforms across AWS and GCP. Strong background in data lakehouse architectures, Airflow-based orchestration, MongoDB, Snowflake, and S3-based data lakes. Known for owning complex data systems end to end - from design and implementation to production stability - and for building reusable, configuration-driven frameworks that improve team productivity and data reliability.

Core Technical Skills

- **Languages:** Python, SQL
- **Cloud Platforms:** AWS (S3, MWAA, ECS, Lambda, CloudWatch, Secrets Manager), GCP (BigQuery, Dataflow, Pub/Sub, Cloud Composer, Cloud Run, Looker Studio)
- **Data Platforms:** MongoDB, Snowflake, BigQuery
- **Data Engineering:** Apache Airflow, Airbyte, Apache Beam, Spark, ETL/ELT Pipelines, Incremental Sync, Data Validation
- **Frameworks & APIs:** FastAPI, REST APIs, OAuth 2.0, JMESPath
- **DevOps & Tooling:** Docker, Terraform, Git, CI/CD workflows, Cursor AI (AI-assisted development)
- **Monitoring:** Datadog, Cloud Monitoring

Professional Experience

Senior Data Engineer - Sequoia

Sep 2024 - Present

- Core contributor to the Sequoia Cloud Data Exchange / Lakehouse Platform, a multi-tenant data platform ingesting and standardizing HR and ATS data from 20+ third-party systems for 50+ customer organizations.
- Designed and implemented an ATS incremental sync system by enhancing the Generic Airbyte Sync framework to support hourly syncs, time-based incremental extraction, state lookback handling, and S3 partitioning, reducing data freshness from daily to near-hourly while safely merging incremental and historical data across candidates, applications, and jobs.
- Led end-to-end ATS integrations, including Workable ATS (custom API extraction, JMESPath mappings, CDM generation) and Lever ATS migration from legacy ingestion to Airbyte, implementing OAuth-based authentication, incremental syncs, and batch-optimized processing.
- Designed and built a reusable, configuration-driven MongoDB export framework to ingest curated platform data into AWS S3 and Snowflake, paired with dynamic Apache Airflow orchestration for configuration-driven execution, parallelism, idempotent re-runs, and automated downstream DAG triggering, enabling fully end-to-end pipeline automation without per-entity custom code.
- Improved platform scalability and reliability by introducing batch controls, volume thresholds, distributed locking, and graceful failure handling, ensuring stable processing of large, high-cardinality datasets in production.
- Drove platform reliability and code quality improvements, refactoring complex processing logic, resolving large-scale linting and stability issues, and stabilizing production pipelines with no regression impact.
- Collaborated closely with platform, analytics, and operations teams to align on ingestion standards, data models, monitoring practices, and production readiness, while documenting frameworks to enable faster onboarding and independent extension.

Senior Associate - Cloud Data Engineer, BlackBuck Insights

Dec 2021 - May 2024

- Worked as a Cloud Data Engineer for a US-based logistics and parcel shipping client, supporting analytics and ML-driven use cases.
- Developed and maintained 50+ production ETL pipelines using Apache Airflow (Cloud Composer) for ingestion from Snowflake and GCS into BigQuery.
- Improved pipeline performance and reliability, reducing average runtimes by ~30%.
- Integrated Vertex AI models into Airflow DAGs to enable demand forecasting and advanced analytics directly on warehouse data.
- Built and maintained Docker-based ML deployments on Cloud Run, reducing model release cycles by ~40%.
- Developed and operated 10+ FastAPI microservices exposing ML outputs to downstream applications with improved latency.
- Implemented Datadog-based monitoring and alerting across pipelines and services, supporting 99.9% SLA compliance.
- Created reusable DAG templates and onboarding documentation, reducing setup time for new tenants by ~40%.

Associate Consultant - Cloud Data Engineer, Virtusa

Jul 2019 - Dec 2021

- Supported ~1,000 GCP-based production data pipelines as part of Google's External Workforce for a large mining client.
- Played a key role in migrating ~100 TB of data from on-prem systems to BigQuery using Apache Beam and Dataflow.
- Built an automated BigQuery disaster recovery solution for schema and data backups, reducing recovery time from hours to minutes.
- Designed and implemented IoT data pipelines for real-time sensor ingestion and analytics, integrated with Virtusa's Cognisense platform.
- Authored Terraform modules to provision GCS, Compute Engine, and BigQuery resources aligned with CIS compliance standards.
- Built Looker Studio dashboards to provide visibility into data platform health and KPIs for senior stakeholders.
- Collaborated with Virtusa X-Labs on applying machine learning to healthcare analytics, contributing to a CNN-based pneumonia detection prototype.

Education & Certifications

Education: B.Tech - Chemical Science & Technology, Indian Institute of Technology, Guwahati (2019)

Certifications: GCP Professional Data Engineer | GCP Professional Cloud Architect

Awards & Recognition

- Sequoia BackPack Award (2025) - Impactful contributions to enterprise data platform initiatives
- BBI Core Value Award (Q3 2023) - Collaboration and reliability
- Virtusa Top Talent (H2 FY21) & Top Performer Bonus (2020-21)
- Virtusa V-Energiser Award (May 2021)